

HIGH FRONTIER

4 ALL | BY PHIL EKLUND



READ ME FIRST

BY JON MANKER & SIMON NG



Don't Panic¹

...and welcome to the *High Frontier* universe. Since its first edition in 2010, *High Frontier* has been the definitive science-based adventure game of space exploration and development. Based on real science & technology (and designed by a bona fide rocket scientist), it has been fine-tuned by a team of enthusiasts for a decade to be playable, comprehensive, and packed with the latest developments in engineering and planetary science. This document provides a progressive learning path, a full list of components, and advice on how to engage with the complexities of *High Frontier*. It also explains how this edition differs from *High Frontier 3*, lists reference material, and gives credit where credit is due.



Where do I begin?

To tell the story of how great a game can be

The game of space that is older than the sea

The simple truth about you playing it with me

Where do I start?

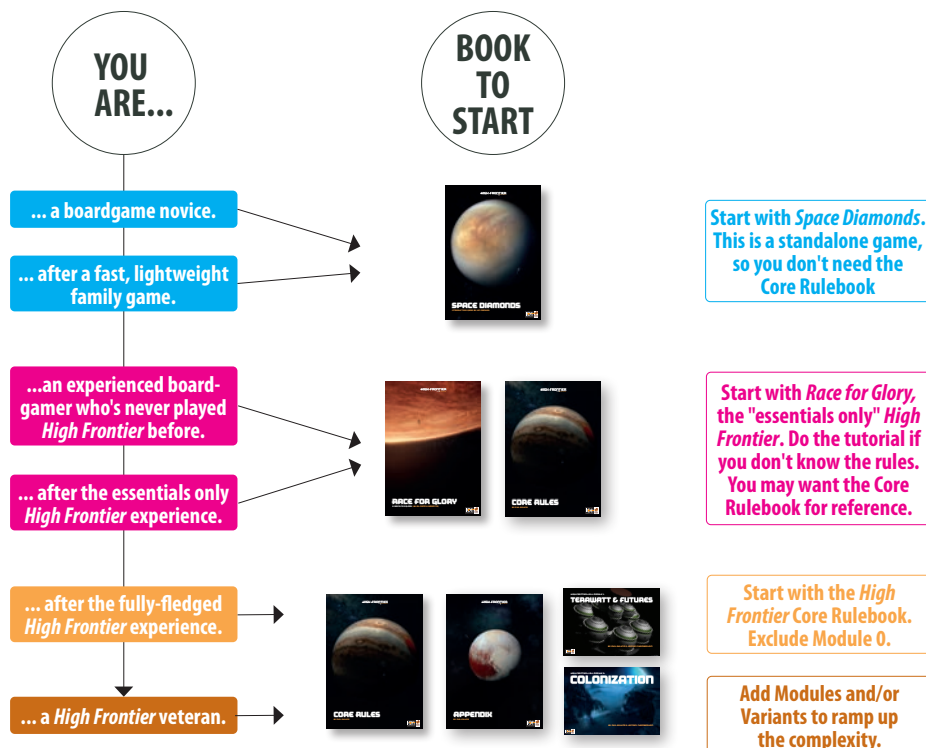
A Progressive Learning Path

High Frontier has a reputation for being difficult to learn. Consistent with the *High Frontier 4 All* name, we've designed a progressive learning path to ease your way through the *High Frontier* universe. Depending on the complexity level your group is comfortable with, this path includes several books:

Space Diamonds is a lightweight, family-friendly race through the solar system. It is a standalone game that uses the standard map from *High Frontier*, but has its own set of rules, rocket/sail and mission cards, and discovery chits.

Race for Glory is the "essentials only" *High Frontier*, equivalent to "Basic" in previous versions of *High Frontier*. It is not a standalone rulebook, but instead references the Core rulebook when helpful. It uses its own set of two-sided cards and includes a comprehensive tutorial.

Core Rulebook is the baseline version of *High Frontier*. It adds rules for supports, thermal control, power requirements, solar flares & Radiation Belts, Glitches, hazardous operations, events, and more. It uses its own set of two-sided cards. Here is an outline of your progressive learning path:



¹ FOOTNOTES. The cover of *The Hitchhiker's Guide to the Galaxy* (by Douglas Adams) has the admonition "Don't Panic". Maybe we should include these "large friendly letters" on our cover as well, in case things look "insanely complicated". In any case, welcome to the first footnote of the game! Footnotes contain background material used to create the rules, and contain no rules necessary to play *High Frontier*.

Components – HF4 Core Box

1 Box

5 Rules Booklets* (*Read Me First, Space Diamonds, Race for Glory, Core Rules, Appendix*)

2 Player Aids*

1 Mapboard

2 Placards: Sol Sunspot Cycle/RfG Sunspot Cycle and Political Assembly/RfG Player Aid

5 Playmats with Fuel Strips*

35 wooden domes (7 each in 5 player colors) for Colonies

35 wooden cubes (7 each in 5 player colors) for Factories

1 gold wooden star for Active Law (Module 0 - Politics)

45 plastic disks (9 each in 5 player colors) for Claims

10 plastic rockets (2 each in 5 player colors). For rocket stack location and net thrust.

80 blue plastic beads for Aqua and water FT

40 transparent red plastic disks for Busted Sites, Seniority & Glitches*

1 black six-sided dice (1d6) used for prospecting, hazards, flares and glitches.

1 Punchout board* (95 Discovery Chits, 10 Outpost markers, 20 Dry and Wet Rocket Mass Chits, 9 Glory and 4 Heroism Chits)

155 Cards (35 Space Diamonds, 36 Race For Glory, 77 Core patents, 6 Crew, 1 map overlay)

10 Ziplock Bags

* Components included in HF Upgrade kit.

High Frontier Learning FAQ

Do I Need to Immediately Read the Core Rulebook?

No. Follow the progressive learning path above to see what you should read to start your adventure. If you start with *Race for Glory*, keep the Core rulebook handy for reference. If you start with *Space Diamonds*, you will not need the Core rulebook at all because *Space Diamonds* uses its own set of rules.

What Other Teaching Materials are Available?

- **A Glossary** is found at the end of the Core rulebook. Capitalized terms in the books are defined here. Useful for any version of *High Frontier*.
- **The Player Aid** defines icons and summarizes game procedures, and is useful for any version of *High Frontier*.
- **Heavy Cardboard Video Tutorial.** The YouTube channel Heavy Cardboard has produced a video tutorial of this game. Links are found in our webshop www.ionsmg.com under the section "Video". Either use this QR-code (right), or just google "High Frontier Heavy Cardboard".
- **Tabletop Simulator, Vassal & Cyberboard.** Three tools are available to allow you to play *High Frontier 4 All* digitally: the Tabletop Simulator found on Steam, a Vassal module located at www.vassalengine.org (the game is found among the modules) and a Cyberboard version found at <http://www.garycki.com/highfrontier.html>.



QR-code for Heavy Cardboard HF4 Playthrough.

How To Use the Modules?

Modules extend your *High Frontier* experience beyond the Core game, combined in any way you'd like,

- **Module 0 – Politics** is included in the Core box, and includes Faction delegates who vote on rule changing laws and provide alternate endgame scoring conditions (**O**).
- **Module 1 – Terawatt** adds thrusters in the trillions of watts, and flexible high-technology (and low-technology) Freighters. Sold separately.
- **Module 2 – Colonization** adds orbital habitats and oddball space Colonists who may come into conflict with Earth. Sold separately.
- **Futures** are mega-engineering missions involving the farthest reaches of the map, and is possible if you have both Modules 1 and 2.

TIP MARGINALIA: Hints, tips, and reminders found in the rule margins.

Other Variants?

- **Appendix.** This includes: scenarios (**V**); strategy tips (**W**); experimental nerdy rules (**X**); essays about the thoughts, ideas and the vision behind the game (**Y**); and finally a card-by-card description of each patent's functions and specifications (**Z**).
- **Bios:Earth Campaign** are rules to integrate *High Frontier* into a grand campaign that plays through all three Bios-games (*Bios:Genesis*, *Bios:Megafauna 2*, and *Bios:Origins*) before starting your game of *High Frontier* using whatever creatures have evolved from the primordial soup in the course of the three prior games. You will use the Core rules and either **V7** or **V8** in the appendix (depending upon your trajectory through the *Bios* games).

Changes from High Frontier 3

Introduction

- **Water Tanks** in LEO are now referred to as *Aquas* (A1). Aquas continue to serve the dual purpose of currency and as a source of Fuel for Rockets.
- **Endgame** for the Core Game is after 48 turns (A1a).
- **Race for Glory** replaces the Basic Game as the essential rules of *High Frontier 4 All*. The *Race to Mars* tutorial teaches the *Race for Glory* game and much of the Core rules (A3).

Components and Anatomies

- **Dry Mass & Wet Mass Indicators** are now colored chits (B1).
- **Number of Factory cubes** increased from 6 to 7 (B1). Colony Domes are no longer limited to what is supplied.
- **Ventures** have been removed. Glory cards have been replaced by 9 *glory chits*. Glory chits are placed on the map and can be loaded into Spacecraft (B3).
- **Exploitation Tracks** reads 8/5/4 victory points, rather than 10/8/6, and introduces the **H** Spectral Type (B5).
- **Burn Space** is a magenta-colored map icon (B7c).
- **Cycler** is removed from Core. It is now only a Bernal Ability.

Setup

- **Senility** as a term is now replaced with the term *Seniority* (C1).
- **Start Crew**. Players secretly choose which side of their Crew card to use (C2).
- **Each Crew** begins in LEO, goes to LEO if *Decommissioned* (E7), and does not need to be Decommissioned for *glories* (La).
- **1st Player & Sunspot Cube** have been introduced. 1st Player is chosen based on the clout of their Crew (C8a).

Yearly Play Sequence

- **Each Year** is now divided into two distinct phases: the *player turn phase* (D1) and the *advance sunspot cycle phase* (D2).

Hands & Stacks

- **Faction privileges** redone so that all 12 Factions have unique privileges.

Spacecraft Mass & Fuel Management

- **Obsolete Terms** include Water Tanks (WTs), Fungible Fuel Tanks (FFT), and Depots. FFTs are now simply Fuel Tanks (FTs). FTs no longer have rad-hardness (B2j). See F1.
- **Thrust & Mass** are calculated at the start of movement (F2,3).

Free Actions

- **Dirt Refueling** is now a free action (*cargo transfer*, G1c), and gives a different amount of fuel for Crew thrusters than for other dirt thrusters.
- **Jettison** is now no longer allowed during movement (G) and jettisoned material will not remain on the map (G1f, G1g).
- **Dirt FTs** (fuel carried as cargo) no longer exist (G2).
- **Colonies** are permanent. Terminology has been changed from Space Colony (G3).
- **Tokens** are *Discarded* rather than *Decommissioned* (G6).
- **Glitch Repair Free Action** has been added (G7).

Movement

- **Thrust Movement Points** (TMPs) has been removed as a term.
- **Spacecraft** must activate an operational thruster in order to move (H2).
- **A Thrusters** no longer needs Fuel to be *activated* (H2a).
- **Spacecraft** with a net thrust of zero or less are allowed to *coast* (H2b).
- **Push Factory & Powersat** have been combined into a single concept (H3d).
- **Bonus Pivot** added to some cards (e.g. Sails) (H4c).
- **Burn** is spending an amount of Fuel equal to *fuel consumption* (H5a).
- **Synodic Comet Sites** cannot be entered or exited by a Spacecraft unless the Sunspot Cube is in the matching-colored season (H6a, B7h).
- **Atmospheric Factories** can use *factory-assist* (H6c).
- **Factory-assist** is not a Hazard if you have Powersat (H6c).
- **Hazard Rolls** occur on a roll of "1" and result in Stack *Decommission*. Radiation Hazards have been renamed Radiation Belts.
- **Bonus Burns** are available only with an operational and activated *thrust triangle* (H8a).

TIP: Rules references for Modules are prefixed with the Module number (e.g. 2C1a for Module 2). The prefix "i" (introductory) indicates *Space Diamonds*, and the prefix "t" (tutorial) indicates *Race for Glory*.

TIP: Some rules that were experimental in HF3 are now standard.

Operations

- **Income Operations** now earn a player 1 Aqua (I1).
- **Free Market Operations** earn 3 Aqua for White-Sided Cards (I3a).
- **ISRU Refuel Ops** for isotope Fuel are no longer possible (I5a).
- **Factory-Refuel** generates 7 FTs rather than 8 (I5b).
- **Raygun Prospecting Restrictions** on Synodic Comet Sites has been removed (but the map has been altered to achieve the same effect).
- **Prospecting Luna** requires permission of the 1st player (I6.1).
- **Prospecting the Kreutz Sungrazer** is automatically successful (I6.3).
- **Basic Game Freighters** are now replaced by the *delivery operation* (I9).

Supports

- **Afterburn Cooling** can only be applied to thruster Stack cards, and only during movement (J4d).
- **New Rounding Rules** for some specific fuel consumption instances (J5c).
- **Supports** in the support chain that count as movement-modifying supports has changed (J5d).

Sunspot Cycle Events

- **Blue Event** has been changed to *anarchy* (K2e).
- **Glitch** has been added as a new event. Rules to cover this new addition are: *Glitch Event* (K2b), *Glitch Rolls* caused by *Glitch Triggers* (H8g, G1h, I5, I6, I7, I9), and *Glitch Repair Free Action* (G7, N8). Glitches are carried on a Stack as red disks. Glitches are triggered by *cargo transfer free actions*, *prospect operations*, *site refuel operations*, *industrialize operations*, and *flyby maneuvers*.
- **Pad Explosion / Space Debris Events** do not affect Crew, Black-side Cards, Purple-side Cards, Colonists, and Bernal's.

TIP: Glitches are no longer a bolts from the blue that cripple everyone simultaneously. Rather they are potential failures requiring risk management.

Glories

- **Chits** now stored in the Heliocentric Zones to be picked up by Humans.
- **Heroism** expanded to 4 categories as listed on the map (Lc).

TIP: Heroism rewards suffering devastating events, loss of life, helping a stricken opponent, or newbie miscalculations.

Game End & Scoring

- **Token VP** includes every wooden or plastic Token on the map of your color.
- **Glory Chits** are worth 1 VP each if claimed, and various VP each if returned to LEO (M2b).²
- **Colonies** are worth variable victory points (M2b).
- **Factories** on special Sites (e.g. Astrobiology, Submarine) are not worth extra VP.

Negotiation

- **Negotiation Rules** have been restructured. The ability to negotiate a *Glitch Repair* has been added (N8).

Map Compared to HF3

- **Science & TNO Science** have been replaced by Astrobiology. "Lost Astrobiology" in the list of Sites below includes Sites that had Science or TNO Science in *High Frontier 3* but do not have Astrobiology in this game.
- **Atmospheric Sites** indicated by a new icon.
- **Route Yellow** goes to Lutetia instead of Ceres.
- **Route Blue** goes to Ceres instead of Comet Enke.
- **Signposts** show Burns but not Hazards.
- **Burn Spaces & Intersections** have been added and removed, including:
 - The purple route has fewer intersections (more bridges) between Luna and the Mars Heliocentric Zone.
 - Earth & Venus flyby approaches are changed.
 - Jupiter flyby approaches and the route to Saturn are changed.
 - The Mars North Pole routes are changed.
 - Uranus / Neptune zone paths are significantly changed.
 - The *half-burn* near Ceres is no longer a Hazard.
 - Many Burn Spaces that were also Hazards have been split into two Spaces.
 - Hazard Spaces next to Synodic Comet Sites are also synodic.
- **Earth Space Elevator** from *High Frontier 3* no longer exists.
- **Eight New Space Elevators** added to the map (in addition to Mars-Phobos) and may be built using the *Freighter rules in Module 1* (I89).
- **Many Aerobrake Hazards** have solid entry routes to emphasize that Spacecraft may move into the Aerobrake Hazard and then exit without landing on a later Turn. This is mainly useful with the *air-eater refuel op* (I5c).
- **Numerous Bernal Home Orbits added** (2B3).
- **Heliocentric Map Locator** (using clock hours) can be used to describe locations on the map (B7I).
- **Specific Site Changes** are listed in the table on next page:

TIP: Distance to the outer solar system is now a greater and more realistic number of Burns and Turns.

TIP: Blue font indicates rules found in the Modules rather than in Core.

² LEO stands for Low Earth Orbit. The International Space Station is in LEO, with an altitude of 400 km. The orbital speed at this altitude is 7.5 km/sec, equivalent to 3 Burns in the game and fast enough to circle the Earth every 92 minutes.

Site Name	Size	Spectral Type	Hydration	Science/TNO/Push/Astrobiology/Submarine	Helio-centric Zone	HF4 Changes
Mercury Discovery Rupes	10	S	0	Push Factory	Mercury ♀	Lost Astrobiology
Hathor	1	S	1	Push Factory	Venus ♀	Gained Push Factory
Apophis	1	S	0	Push Factory	Earth ⊕	Gained Push Factory
Bennu	1	C	2	Push Factory	Earth ⊕	New site
Cruithne	1	M	1	Push Factory	Earth ⊕	Gained Push Factory
Icarus	1	S	4	Push Factory, Astrobiology	Earth ⊕	Gained Push Factory, Hydration increased
Itokawa	1	S	0	Push Factory	Earth ⊕	Gained Push Factory
Khufu	1	S	2	Push Factory	Earth ⊕	Gained Push Factory
Luna Aristarchus Plateau	9	H	0		Earth ⊕	Changed Spectral Type
Luna Crater Shackleton Polar Rim	9	S	1		Earth ⊕	Hydration reduced
Mjölñir	1	S	1	Push Factory	Earth ⊕	Gained Push Factory
Phaethon	1	H	4	Push Factory. Astrobiology	Earth ⊕	Changed Spectral Type, Gained Push Factory
YORP	1	S	0	Push Factory	Earth ⊕	Gained Push Factory
Deimos	1	D	2		Mars ♂	Changed Spectral Type, Hydration reduced
Eureka	1	D	0	Astrobiology	Mars ♂	Changed Spectral Type, Always Astrobiology
Badenia	2	D	2		Ceres ♀	Changed Spectral Type
Churyumov-Gerasimenko	1	C	4	Astrobiology	Ceres ♀	New site
Comet Holmes	1	D	4	Astrobiology	Ceres ♀	Synodic (Red)
Comet Wilson-Harrington	1	D	4	Astrobiology	Ceres ♀	No longer synodic, gained Astrobiology
Friederike	2	D	2		Ceres ♀	Changed Spectral Type
Lacrimosa	3	D	0		Ceres ♀	Changed Spectral Type
Polana	2	D	1		Ceres ♀	Changed Spectral Type
Bee-Zed	1	H	4	Astrobiology	Jupiter ♃	New site
Europa Subsurface Ocean	8	C	4	Subsurface Ocean	Jupiter ♃	Changed to Subsurface Ocean
Skamandrios moonlet	1	D	3		Jupiter ♃	Name change (from “moonlet”)
Echeclus	2	D	4	Astrobiology	Saturn ♄	New site
Enceladus	5	M	4	Subsurface Ocean	Saturn ♄	Changed to Subsurface Ocean
Epimetheus	2	C	4	Astrobiology	Saturn ♄	Gained Astrobiology
Janus	2	C	4	Astrobiology	Saturn ♄	Gained Astrobiology
Kreutz Sungrazer	1	H	4	Astrobiology	Saturn ♄	Changed Spectral Type, Prospecting Size rolls always succeed
Methone	1	D	4	Subsurface Ocean	Saturn ♄	Changed to Subsurface Ocean
Pandora	2	C	4	Astrobiology	Saturn ♄	Gained Astrobiology

Site Name	Size	Spectral Type	Hydration	Science/TNO/Push/Astrobiology/Submarine	Helio-centric Zone	HF4 Changes
Prometheus	2	C	4	Astrobiology	Saturn ♄	Gained Astrobiology
Saturn Aerostat	11	H	0	Astrobiology	Saturn ♄	Changed Spectral Type, Lost Astrobiology
Tethys	5	C	4	Astrobiology	Saturn ♄	Gained Astrobiology
Titan Aerostat	8	C	0	Astrobiology, Atmospheric	Saturn ♄	New site
Titan Kraken Mare	9	V	3	Subsurface Ocean, atmospheric	Saturn ♄	Changed to Subsurface Ocean
Ariel	6	V	4	Subsurface Ocean	Uranus ♅	Changed to Subsurface Ocean
Belinda	2	V	4	Astrobiology	Uranus ♅	Gained Astrobiology
Chariklo	4	D	4	Astrobiology	Uranus ♅	Changed Spectral Type
Comet Halley	1	H	4	Astrobiology	Uranus ♅	Changed Spectral Type
Cordelia	1	C	4	Astrobiology	Uranus ♅	Gained Astrobiology
Hylonome	3	D	4		Uranus ♅	Lost Astrobiology
Juliet	2	V	4	Astrobiology	Uranus ♅	Gained Astrobiology
Oberon	7	D	4	Subsurface Ocean	Uranus ♅	Lost Astrobiology
Portia	3	V	4	Astrobiology	Uranus ♅	Gained Astrobiology
Titania	7	D	4	Subsurface Ocean	Uranus ♅	Changed Spectral Type, Lost Astrobiology
Uranus Aerostat	11	H	1	Subsurface Ocean , Atmospheric	Uranus ♅	Changed Spectral Type, Changed to Subsurface Ocean, Gained Atmospheric
1999OY	2	V	4	Astrobiology	Neptune ♆	Changed Spectral Type
2002TX	4	D	4	Astrobiology	Neptune ♆	Changed Spectral Type
Arrokoth	1	D	4	Astrobiology	Neptune ♆	New site
Charon	6	D	4	Astrobiology	Neptune ♆	Changed Spectral Type
Comet McNaught	1	H	4	Astrobiology	Neptune ♆	Changed Spectral Type, No longer synodic
Larissa	3	D	3	Astrobiology	Neptune ♆	Gained Astrobiology
Neptune Aerostat	11	H	2	Astrobiology, Atmospheric	Neptune ♆	Changed Spectral Type, Gained Atmospheric
Pluto	7	D	4	Subsurface Ocean	Neptune ♆	Changed Spectral Type, Lost Astrobiology, Gained Aerobrake route
Proteus	4	D	3		Neptune ♆	Lost Astrobiology
Quaoar	7	S	4	Subsurface Ocean	Neptune ♆	Changed to Subsurface Ocean
Sao	1	D	4	Astrobiology	Neptune ♆	Gained Astrobiology
Sedna	6	H	4	Subsurface Ocean	Neptune ♆	Changed Spectral Type, Changed to Subsurface Ocean
Triton Mahilani Plume	8	M	4	Subsurface Ocean	Neptune ♆	Lost Astrobiology, Gained Aerobrake route
Triton Tuenela Plantia	8	C	4	Astrobiology	Neptune ♆	Lost Subsurface Ocean, Gained Aerobrake route



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